

DYNAMICS OF CAR-LIKE ROBOTS WITH TRAILERS, APPLICATION TO SKID AND JACK-KNIFE

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Abstract—In this paper, we present a dynamic model for simulating a car-like robot with trailer taking into account suspension system and tires deformations. The model describing the wheel/ground contact is a linear model and the joint tractor/trailer is modeled as a spherical-joint. The obtained results are compared with those given by the Pc-Crash software. We noted a good correlation for various scenarios. Furthermore, the proposed model has been successfully applied to study the unbalanced braking in a straight line and the jack-knife problem.

Keywords— car-like robot, tractor, trailer, dynamics, simulation, skid, jack-knife.

I. INTRODUCTION

In recent decades, research on dynamics of road transport vehicles and autonomous vehicles was the subject of special attention and interest of more and more laboratories, researchers and industrial suppliers. Particularly, mobile robots is the focus of much research because of their increasingly use in dangerous and inaccessible environments, for civilian or military applications. These problems are far from being easy to solve. The difficulty lies in apprehending the overall dynamic behavior of mobile robots particularly in aggressive situations involving braking, abrupt steering, skidding Indeed, there is an urgent need for simulation tools enabling the study of vehicles in normal and extreme situations in order to develop appropriate control laws and trajectory planning strategies.

The objective of the present work is to propose a general dynamic model that allows the simulation of car-like robots dynamics with trailers. Moreover, we illustrate the exploitation of this model for the investigation of severe scenarios involving skid or reversal events and jack-knife problems (situations present in road accidents). The proposed approach for the implementation of this model is based on versatile methods capable of handling different possible behaviors with minimal changes. The wheels/ground contact is handled using a linear model ([1], [2], [3] and [4]) and the joint tractor/trailer is modeled as a spherical-joint. The simulation results obtained

by the developed model are validated with those obtained by the Pc-Crash® software.

II. DYNAMIC MODELING

A. Geometric description of the system

In the present study, the structure of the considered mobile robots are composed of a tractor and a trailer (Fig. 1) which are supposed rigid bodies fixed to the chassis and suspended by the suspension system composed of a spring and damper (k_i stiffness and damping coefficient b_i). The articulation between the tractor and the trailer is modeled as a spherical-joint. Unsuspended masses are negligible compared to the suspended masses. In addition, the caster, toe and tilt angles of the kingpin are not considered. The anti-roll bar is not modeled. Traction is always reserved for the tractor, also the braking is applied to all wheels.

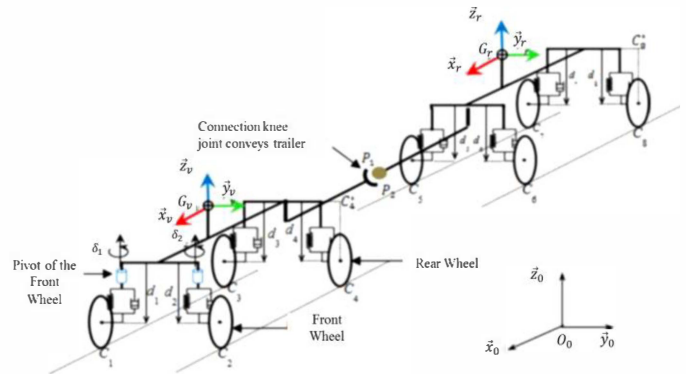


Fig. 1. Model of a car-like robot with trailer

The system is moving on a road to which is attached a frame $\mathcal{R}_0 = (O_0, \vec{x}_0, \vec{y}_0, \vec{z}_0)$. Let $\mathcal{R}_v = (G_v, \vec{x}_v, \vec{y}_v, \vec{z}_v)$ be the mobile frame linked to the tractor chassis, G_v the original center of gravity of the tractor. The axis $\vec{x}_v$ is perpendicular to the plane of longitudinal symmetry of the tractor, facing the left side, the axis is perpendicular to $\vec{y}_v$ and directed towards the

front of the tractor. The moving frame $\mathcal{R}_r = (\vec{G}_r, \vec{x}_r, \vec{y}_r, \vec{z}_r)$ attached to the chassis of the trailer is defined in the same way as the tractor frame.

The configuration of the tractor frame versus the fixed reference $\mathcal{H}_0$ is defined by six parameters: three Cartesian coordinates (x_1, y_1, z_1) of the center of gravity and the three angles (φ_1 : roll, φ_2 : pitching and φ_3 : yaw) orientation of the $\mathcal{H}_v$ with respect to reference $\mathcal{H}_0$. The rotation matrix of $\mathcal{H}_v$ relatively to $\mathcal{H}_0$ is:

$${}^0P_v = \begin{bmatrix} c_2c_3 & s_1s_2c_3 - c_1s_3 & c_1s_2c_3 + s_1s_3 \\ c_2s_3 & s_1s_2s_3 + c_1c_3 & c_1s_2s_3 - s_1c_3 \\ -s_2 & s_1c_2 & c_1c_2 \end{bmatrix} \quad (1)$$

Similarly, the situation of the frame $\mathcal{H}_r$ trailer relative to the $\mathcal{H}_0$ is defined by six parameters: $x_2, y_2, z_2, \varphi_4, \varphi_5$ and φ_6 . The rotation matrix of $\mathcal{H}_r / \mathcal{H}_0$ is as follows:

$${}^0P_r = \begin{bmatrix} c_5c_6 & s_4s_5c_6 - c_4s_6 & c_4s_5c_6 + s_4s_6 \\ c_5s_6 & s_4s_5s_6 + c_4c_6 & c_4s_5s_6 - s_4c_6 \\ -s_5 & s_4c_5 & c_4c_5 \end{bmatrix} \quad (2)$$

With: $c_i = \cos(\varphi_i), s_i = \sin(\varphi_i), i = 1..8$.

The kinematic wrenches of the tractor frame and the trailer frame relative to the reference, expressed respectively in the tractor frame and in the trailer frame, are given by the following relationships:

$$\begin{aligned} {}^v\vec{V}_{Gv} = {}^vP_0 \begin{pmatrix} \dot{x}_1 \\ \dot{y}_1 \\ \dot{z}_1 \end{pmatrix} \text{ and } {}^v\vec{w}_v = \begin{pmatrix} w_{xv} \\ w_{yv} \\ w_{zv} \end{pmatrix} &= \begin{bmatrix} 1 & 0 & -s_2 \\ 0 & c_1 & s_1c_2 \\ 0 & -s_1 & c_1c_2 \end{bmatrix} \begin{pmatrix} \dot{\varphi}_1 \\ \dot{\varphi}_2 \\ \dot{\varphi}_3 \end{pmatrix} \\ {}^r\vec{V}_{Gr} = {}^rP_0 \begin{pmatrix} \dot{x}_2 \\ \dot{y}_2 \\ \dot{z}_2 \end{pmatrix} \text{ and } {}^r\vec{w}_r = \begin{pmatrix} w_{xr} \\ w_{yr} \\ w_{zr} \end{pmatrix} &= \begin{bmatrix} 1 & 0 & -s_5 \\ 0 & c_4 & s_4c_5 \\ 0 & -s_4 & c_4c_5 \end{bmatrix} \begin{pmatrix} \dot{\varphi}_4 \\ \dot{\varphi}_5 \\ \dot{\varphi}_6 \end{pmatrix} \end{aligned}$$

In order to take into account the efforts of wheels/ground interaction, we attach to each wheel, a frame $\mathcal{H}_i = (\vec{G}_i, \vec{e}_{xi}, \vec{e}_{yi}, \vec{e}_{zi})$ placed at the wheel/ground contact point. The unit vectors of $\mathcal{H}_i$ are defined, according to the normal $\vec{n}_i$ of the wheel contact i with the ground plane and its steering angle, by the following expressions:

$$\begin{aligned} \text{For } i = 1..4 \quad & \begin{cases} {}^{0-}\vec{e}_{zi} = {}^{0-}\vec{n}_i \\ {}^{0-}\vec{e}_{xi} = \frac{({}^0P_v \cdot {}^{v-}\vec{v}_i) \wedge {}^{0-}\vec{e}_{zi}}{|({}^0P_v \cdot {}^{v-}\vec{v}_i) \wedge {}^{0-}\vec{e}_{zi}|} \\ {}^{0-}\vec{e}_{yi} = {}^{0-}\vec{e}_{zi} \wedge {}^{0-}\vec{e}_{xi} \end{cases} \\ \text{For } i = 5..8 \quad & \begin{cases} {}^{0-}\vec{e}_{zi} = {}^{0-}\vec{n}_i \\ {}^{0-}\vec{e}_{xi} = \frac{({}^0P_r \cdot {}^{r-}\vec{v}_i) \wedge {}^{0-}\vec{e}_{zi}}{|({}^0P_r \cdot {}^{r-}\vec{v}_i) \wedge {}^{0-}\vec{e}_{zi}|} \\ {}^{0-}\vec{e}_{yi} = {}^{0-}\vec{e}_{zi} \wedge {}^{0-}\vec{e}_{xi} \end{cases} \end{aligned} \quad (3)$$

With $\vec{n}_i$ the vector normal to the wheel plane. This vector is defined by ${}^{v-}\vec{v}_i = [-\sin \delta_i \quad \cos \delta_i \quad 0]^T$ for the front wheels of the tractor and ${}^{r-}\vec{v}_i = [0 \quad 1 \quad 0]^T$ for the other wheels, ${}^{r-}\vec{v}_i = [0 \quad 1 \quad 0]^T$ for all wheels of trailer.

The transition matrix of the wheel/ground interaction frame ($\mathcal{H}_i / \mathcal{H}_0$) is defined by the following relationship:

$${}^0P_i = \begin{bmatrix} {}^{0-}\vec{e}_{xi} & | & {}^{0-}\vec{e}_{yi} & | & {}^{0-}\vec{e}_{zi} \end{bmatrix} \quad \text{for } i = 1..8 \quad (4)$$

In this, the elongations of the springs are modeled by the variables d_i , for $i = 1..8$. The vector of the generalized coordinates $\vec{q}$ contains the kinematic variables of translation and orientation of each robot (tractor and trailer), the variables characterizing the elongation of the springs and the two front wheels steering angles of the tractor δ_1 and δ_2 . The vector $\vec{q}$ is thus defined as follows:

$$\vec{q} = (x_1, y_1, z_1, \varphi_1, \varphi_2, \varphi_3, d_1, d_2, d_3, d_4, x_2, y_2, z_2, \varphi_4, \varphi_5, \varphi_6, d_5, d_6, d_7, d_8, \lambda_x, \lambda_y, \lambda_z, \delta_1, \delta_2)^T$$

B. Constraint equations

The equations of constraints of a car-like robot with trailer are defined by the conditions linked to Ackermann steering system, the contact holding conditions of wheels/ground for both tractor and trailer and the conditions of the tractor/trailer connection. They are developed bellows.

1) Ackermann Conditions

During the movement of a tractor with a trailer along a curve, the steering angles of the tractor are linked to the existing of yaw speed w_{yv} and the radius of curvature of this curve by the following equations:

$$\begin{aligned} \text{if } w_{zv} = 0 \Rightarrow & \begin{cases} \delta_1 = 0 \\ \delta_2 = 0 \end{cases} \\ \text{if } w_{zv} \neq 0 \Rightarrow & \begin{cases} \delta_1 = \tan^{-1}\left(\frac{L_v}{R + L_w/2}\right) \\ \delta_2 = \tan^{-1}\left(\frac{L_v}{R - L_w/2}\right) \end{cases} \end{aligned} \quad (5)$$

With w_{zv} : Angular speed yaw of the tractor; L_v : tractor wheelbase; L_w : rear track of the tractor; V_{zv} : longitudinal speed of the tractor; $R = V_{zv}/w_{zv}$ The radius of curvature calculated on the rear track of the tractor center.

2) Wheels/ground holding contact conditions (WGHCC)

WGHCC of a tractor and a trailer are defined by the following algebraic equations:

a) WGHCC of the tractor

$$\left({}^0\overline{O_0C_i} - {}^0\overline{O_0G_v} - {}^0\overline{G_vC_i^*} + r_w {}^0\overline{n_i} \right) {}^0\overline{z_v} + d_i = 0 \quad \text{for } i=1...4 \quad (6)$$

b) WGHCC of the trailer

$$\left({}^0\overline{O_0C_i} - {}^0\overline{O_0G_r} - {}^0\overline{G_rC_i^*} + r_w {}^0\overline{n_i} \right) {}^0\overline{z_r} + d_i = 0 \quad \text{for } i=5...8 \quad (7)$$

With r_w : the radius of the wheels and $\overline{n_i}$: the normal of the contact plane of the wheel i .

These conditions (6 and 7), derived twice with respect to time, can be expressed as follows:

$$\begin{cases} [J_v] \cdot \ddot{\overline{x_v}} + \overline{B_{cv}} = \vec{0} \\ [J_r] \cdot \ddot{\overline{x_r}} + \overline{B_{cr}} = \vec{0} \end{cases} \quad (8)$$

With: $[J_v]$ and $[J_r]$ are Jacobian matrices inherent to the WGHCC of the tractor and the trailer respectively;

$\overline{B_{cv}}$ and $\overline{B_{cr}}$ are accelerations constraints vectors of WGHCC of the tractor and trailer respectively;

$$\begin{aligned} \ddot{\overline{x_v}} &= [\ddot{x}_1, \ddot{y}_1, \ddot{z}_1, \dot{w}_{xv}, \dot{w}_{yv}, \dot{w}_{zv}, \ddot{d}_1, \ddot{d}_2, \ddot{d}_3, \ddot{d}_4]^T \\ \ddot{\overline{x_r}} &= [\ddot{x}_2, \ddot{y}_2, \ddot{z}_2, \dot{w}_{xr}, \dot{w}_{yr}, \dot{w}_{zr}, \ddot{d}_5, \ddot{d}_6, \ddot{d}_7, \ddot{d}_8]^T \\ [J_v] &= \begin{bmatrix} {}^{0\rightarrow T}n_1 & -{}^{0\rightarrow T}n_1 \left({}^0P_v \left[{}^v\overline{G_vC_1} \right] \right) & -{}^{0\rightarrow T}n_1 {}^{0\rightarrow T}z_v & 0 & 0 & 0 \\ {}^{0\rightarrow T}n_2 & -{}^{0\rightarrow T}n_2 \left({}^0P_v \left[{}^v\overline{G_vC_2} \right] \right) & 0 & -{}^{0\rightarrow T}n_2 {}^{0\rightarrow T}z_v & 0 & 0 \\ {}^{0\rightarrow T}n_3 & -{}^{0\rightarrow T}n_3 \left({}^0P_v \left[{}^v\overline{G_vC_3} \right] \right) & 0 & 0 & -{}^{0\rightarrow T}n_3 {}^{0\rightarrow T}z_v & 0 \\ {}^{0\rightarrow T}n_4 & -{}^{0\rightarrow T}n_4 \left({}^0P_v \left[{}^v\overline{G_vC_4} \right] \right) & 0 & 0 & 0 & -{}^{0\rightarrow T}n_4 {}^{0\rightarrow T}z_v \end{bmatrix} \end{aligned}$$

$$\begin{aligned} [J_r] &= \begin{bmatrix} {}^{0\rightarrow T}n_5 & -{}^{0\rightarrow T}n_5 \left({}^0P_r \left[{}^r\overline{G_rC_1} \right] \right) & -{}^{0\rightarrow T}n_5 {}^{0\rightarrow T}z_r & 0 & 0 & 0 \\ {}^{0\rightarrow T}n_6 & -{}^{0\rightarrow T}n_6 \left({}^0P_r \left[{}^r\overline{G_rC_2} \right] \right) & 0 & -{}^{0\rightarrow T}n_6 {}^{0\rightarrow T}z_r & 0 & 0 \\ {}^{0\rightarrow T}n_7 & -{}^{0\rightarrow T}n_7 \left({}^0P_r \left[{}^r\overline{G_rC_3} \right] \right) & 0 & 0 & -{}^{0\rightarrow T}n_7 {}^{0\rightarrow T}z_r & 0 \\ {}^{0\rightarrow T}n_8 & -{}^{0\rightarrow T}n_8 \left({}^0P_r \left[{}^r\overline{G_rC_4} \right] \right) & 0 & 0 & 0 & -{}^{0\rightarrow T}n_8 {}^{0\rightarrow T}z_r \end{bmatrix} \\ \overline{B_{cv}}[i] &= {}^{0\rightarrow T}n_i \left[{}^0P_v \left(2 \cdot {}^v\overline{w_v} \wedge (-\dot{d}_i \cdot {}^v\overline{z_v}) + {}^v\overline{w_v} \wedge \left({}^v\overline{w_v} \wedge {}^v\overline{G_vC_i} \right) \right) \right] \quad \text{for } i=1...4 \\ \overline{B_{cr}}[i] &= {}^{0\rightarrow T}n_i \left[{}^0P_r \left(2 \cdot {}^r\overline{w_r} \wedge (-\dot{d}_i \cdot {}^r\overline{z_r}) + {}^r\overline{w_r} \wedge \left({}^r\overline{w_r} \wedge {}^r\overline{G_rC_i} \right) \right) \right] \quad \text{for } i=5...8 \\ [{}^0\overline{G_vC_i}] &: \text{The operator of the vector cross product.} \end{aligned}$$

3) Constraints of the tractor / trailer connection

The joint between the tractor and the trailer is modeled as a spherical-joint. The constraints of the connection are defined by the following relationship:

$${}^0\overline{O_0P_1} = {}^0\overline{O_0P_2} \quad (9)$$

The derivation of these constraints with respect to time gives the following acceleration equations:

$$[J_{Lv}] \cdot \ddot{\overline{x_v}} - [J_{Lr}] \cdot \ddot{\overline{x_r}} + \overline{B_{Lv}} - \overline{B_{Lr}} = \vec{0} \quad (10)$$

With: $[J_{Lv}]$ and $[J_{Lr}]$ are Jacobian matrices constraints of the tractor / trailer connection of the tractor and trailer respectively;

$\overline{B_{Lv}}$ and $\overline{B_{Lr}}$ are accelerations constraints vectors of the tractor / trailer connection of the tractor and trailer respectively;

$$\begin{aligned} [J_{Lv}] &= \begin{bmatrix} [I_{3 \times 3}] & \left(-{}^0P_v \left[{}^v\overline{G_vP_1} \right] \right) & [0]_{3 \times 4} \end{bmatrix} \\ [J_{Lr}] &= \begin{bmatrix} [I_{3 \times 3}] & \left(-{}^0P_r \left[{}^r\overline{G_rP_2} \right] \right) & [0]_{3 \times 4} \end{bmatrix} \\ \overline{B_{Lv}} &= {}^0P_v \left({}^v\overline{w_v} \wedge \left({}^v\overline{w_v} \wedge {}^v\overline{G_vP_1} \right) \right) \\ \overline{B_{Lr}} &= {}^0P_r \left({}^r\overline{w_r} \wedge \left({}^r\overline{w_r} \wedge {}^r\overline{G_rP_2} \right) \right) \end{aligned}$$

III. EQUATIONS OF MOTION

The main forces acting on a car-like robot in motion with trailer are: wheels/ground interaction force, the dynamic forces, gravity and the aerodynamic forces. In this model, we consider only the dynamic forces, gravity, and wheel/ground interaction efforts for reasons of simplification.

Interaction Efforts applied to each wheel of the system are represented in the reference related to wheels by the vector ${}^i\overline{F_i} = [\overline{F_{xi}}, \overline{F_{yi}}, \overline{F_{zi}}]^T$. These three components are linked to each other by a set of dependency relations defined by the adopted wheel/ground interaction efforts model. Taking

account of the forces applied on the tractor and trailer, the twelve following equations of motion is obtained as follows:

$$\begin{cases} m_v \begin{pmatrix} \ddot{x}_1 \\ \ddot{y}_1 \\ \ddot{z}_1 \end{pmatrix} = m_v \cdot \vec{g} + \sum_{i=1}^4 {}^0\vec{F}_i - {}^0\vec{\lambda}_{v/r} \\ {}^v\mathbb{I}_{Gv} \cdot \vec{\omega}_v + \vec{\omega}_v \wedge ({}^v\mathbb{I}_{Gv} \cdot \vec{\omega}_v) = \sum_{i=1}^4 {}^v\vec{G}_v \vec{C}_i \wedge {}^v\vec{F}_i - {}^v\vec{G}_v \vec{P}_1 {}^0\vec{\lambda}_{v/r} \end{cases} \quad (11)$$

$$\begin{cases} m_r \begin{pmatrix} \ddot{x}_2 \\ \ddot{y}_2 \\ \ddot{z}_2 \end{pmatrix} = m_r \cdot \vec{g} + \sum_{i=5}^8 {}^0\vec{F}_i + {}^0\vec{\lambda}_{v/r} \\ {}^r\mathbb{I}_{Gr} \cdot \vec{\omega}_r + \vec{\omega}_r \wedge ({}^r\mathbb{I}_{Gr} \cdot \vec{\omega}_r) = \sum_{i=5}^8 {}^r\vec{G}_r \vec{C}_i \wedge {}^r\vec{F}_i + {}^r\vec{G}_r \vec{P}_1 {}^0\vec{\lambda}_{v/r} \end{cases} \quad (12)$$

With $\vec{g}$: the gravitational acceleration; $\vec{\omega}_v$: Angular acceleration of the tractor expressed in $\mathcal{R}_v$ frame; $\vec{\omega}_r$: Angular acceleration of the trailer expressed in $\mathcal{R}_r$ frame; ${}^v\mathbb{I}_{Gv}$ and ${}^r\mathbb{I}_{Gr}$ are tensors of inertia, respectively, of the tractor and the trailer; m_v and m_r : are the masses of the tractor and the trailer; $\vec{\lambda}_{v/r}$: the tensile force applied by the tractor to the trailer.

$${}^v\vec{F}_i = -k_i (d_i - L_{0i}) - b_i \cdot \dot{d}_i \text{ for } i = 1 \dots 4 \quad (13)$$

$${}^r\vec{F}_i = -k_i (d_i - L_{0i}) - b_i \cdot \dot{d}_i \text{ for } i = 5 \dots 8 \quad (14)$$

With k_i and b_i are, respectively, the stiffness and the damping coefficient of the wheel i ; L_{0i} : Initial spring length of the wheel i .

By grouping the various equations of constraints and motion of the two sub systems (tractor & trailer) we obtain the following differential-algebraic system describing the dynamics of our system:

$$[A] \cdot \vec{\ddot{X}} = \vec{B} \quad (15)$$

$$[A] = \begin{bmatrix} [A_v]_{10 \times 10} & [0]_{10 \times 10} & [J_{Lv}]_{10 \times 3} \\ [0]_{10 \times 10} & [A_r]_{10 \times 10} & [-J_{Lr}]_{10 \times 3} \\ [J_{Lv}]_{3 \times 10} & [-J_{Lr}]_{3 \times 10} & [0]_{3 \times 3} \end{bmatrix} \quad \vec{\ddot{X}} = \begin{pmatrix} \ddot{x}_v \\ \ddot{y}_v \\ {}^0\vec{\lambda}_{v/r} \end{pmatrix}$$

$$[A_v] = \begin{bmatrix} [M_v]_{6 \times 10} \\ [J_v]_{4 \times 10} \end{bmatrix}; [A_r] = \begin{bmatrix} [M_r]_{6 \times 10} \\ [J_r]_{4 \times 10} \end{bmatrix}$$

$$[M_v] = \begin{bmatrix} m_v I_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times 4} \\ 0_{3 \times 3} & {}^v\mathbb{I}_{Gv} & 0_{3 \times 4} \end{bmatrix}; [M_r] = \begin{bmatrix} m_r I_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times 4} \\ 0_{3 \times 3} & {}^r\mathbb{I}_{Gr} & 0_{3 \times 4} \end{bmatrix}$$

$$\vec{B} = \begin{pmatrix} \vec{B}_v \\ \vec{B}_r \\ \vec{B}_{Lv} - \vec{B}_{Lr} \end{pmatrix}; \vec{B}_v = \begin{pmatrix} \vec{B}_{mv} \\ -\vec{B}_{cv} \end{pmatrix}; \vec{B}_r = \begin{pmatrix} \vec{B}_{mr} \\ -\vec{B}_{cr} \end{pmatrix}$$

$$\vec{B}_{mv} = \begin{pmatrix} m_v \cdot \vec{g} + \sum_{i=1}^4 ({}^0P_i \cdot {}^i\vec{F}_i) + {}^0P_v \cdot {}^v\vec{F}_{extr} \\ -{}^v\vec{\omega}_v \wedge ({}^v\mathbb{I}_{Gv} \cdot \vec{\omega}_v) + \sum_{i=1}^4 {}^v\vec{G}_v \vec{C}_i \times ({}^0P_i \cdot {}^i\vec{F}_i) + {}^v\vec{M}_{extr} \end{pmatrix}$$

$$\vec{B}_{mr} = \begin{pmatrix} m_r \cdot \vec{g} + \sum_{i=5}^8 ({}^0P_i \cdot {}^i\vec{F}_i) + {}^0P_r \cdot {}^r\vec{F}_{extr} \\ -{}^r\vec{\omega}_r \wedge ({}^r\mathbb{I}_{Gr} \cdot \vec{\omega}_r) + \sum_{i=5}^8 {}^r\vec{G}_r \vec{C}_i \times ({}^0P_i \cdot {}^i\vec{F}_i) + {}^r\vec{M}_{extr} \end{pmatrix}$$

The numerical integration of this system of equations, with the consideration of the initial conditions, describe the dynamic behavior of the car-like robot with trailer for a given time history of acceleration (or braking) and steering data. In this work, we used the Euler method for integrating these differential equations.

IV. RESULTS

To discuss the reliability of obtained results we will validate them with those given by the of Pc-crash® accidents simulator. Then, we will present some examples of applications of the modeling approach for different complex scenarios. In these examples, the applied geometrical and inertial parameters of the tractor with trailer are summarized in Table 1.

TABLE I. GEOMETRICAL AND INERTIAL PARAMETERS OF THE TRACTOR/TRAILER

Parameters	Tractor	Trailer
Mass m	950 Kg	1270 Kg
Moment of inertia I_{xx}	1072 Kg/m ²	1933Kg/m ²
Moment of inertia I_{yy}	1342 Kg/m ²	2417Kg/m ²
Moment of inertia I_{zz}	1342 Kg/m ²	2417Kg/m ²
Height of center of gravity	0.50 m	0.40 m
Front track	1.41 m	1.48 m
Rear track	1.41 m	1.48 m
front wheelbase	1.04 m	0.50 m
Rear wheelbase	1.50 m	0.50 m
Front suspension stiffness	9.97 KN/m	16.529 KN/m
Rear suspension stiffness	6.79 KN/m	16.088 KN/m
Front dumping	897.28 N.s/m	1.652 KN.s/m
Rear dumping	611.01 N.s/m	1.608 KN.s/m
Front camber angle	0.00 °	0.00 °
Rear camber angle	0.00 °	0.00 °
Coefficient of friction	0.70	0.70
Radius wheel	0.25 m	0.25 m
Hinge point position	(-2.66, 0.0, -0.3)	(5.0, 0.0, 0.0)

Scenario 1: Acceleration and deceleration on a straight path:

This scenario is described as follows (Fig.2): i) an acceleration phase for 4 seconds; ii) a constant speed phase for 4 seconds and iii) a deceleration (braking is distributed on the front and rear wheels of the tractor) for 4 seconds. The simulation results by the developed application and those given by PC-Crash are illustrated in fig. 3-5. These figures show a good correlation between the results obtained by the dynamic model developed and those given by the software of Pc-Crash.

Scenario 2: Dynamics of a Vehicle with trailer during the course of a turn:

This scenario is described as follows: i) linear movement for 2 seconds at constant speed $S = 54$ km/h; ii) turning within 100m turn for 8 seconds. The obtained simulation results are illustrated in figures 6 to 10. The curves plotted in these figures show a very satisfactory correlation between the results of our developed model and those given by Pc-Crash.

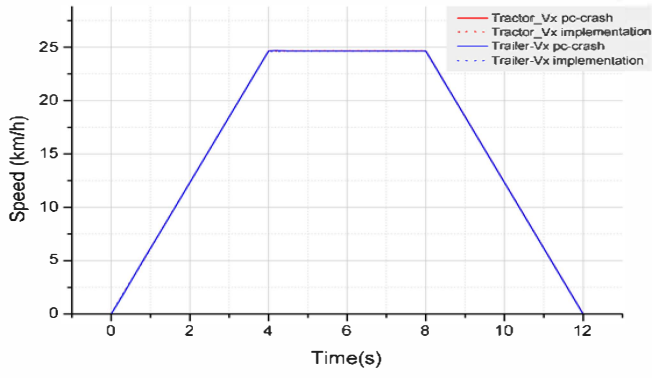


Fig. 2. Speed of the car-like robots (tractor / trailer)

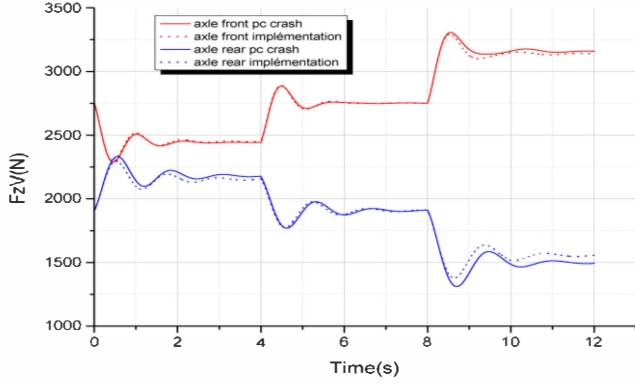


Fig. 3. Vertical Forces wheels / tractor

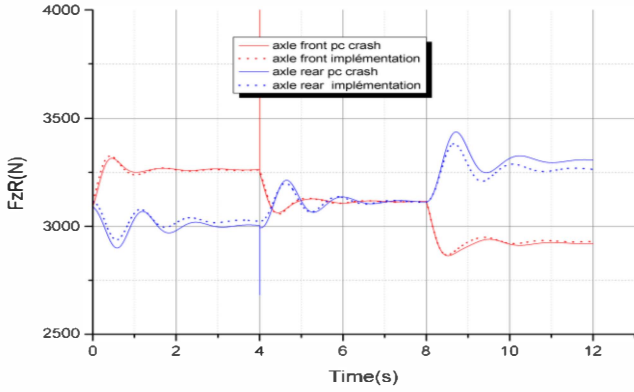


Fig. 4. Vertical Forces wheel / trailer

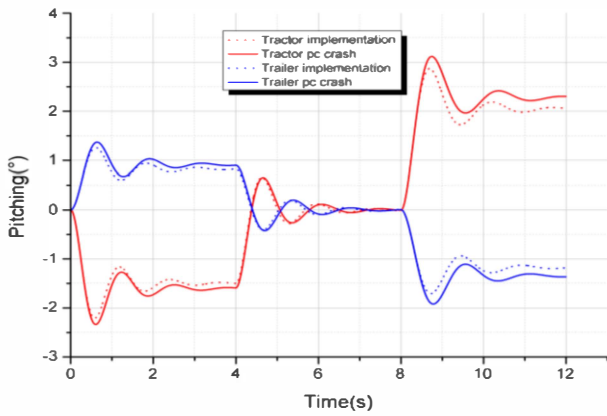


Fig. 5. Pitching angle of the car-like robot

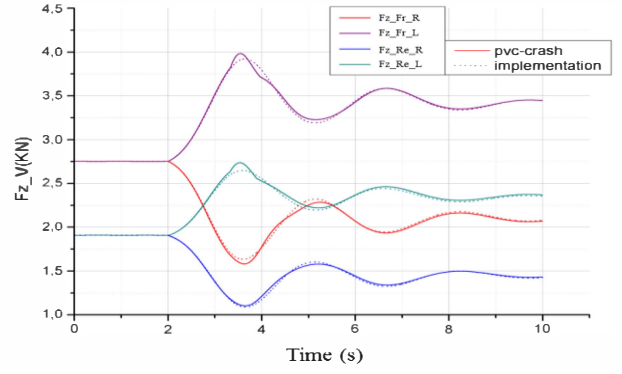


Fig. 6. Vertical Forces wheels / tractor

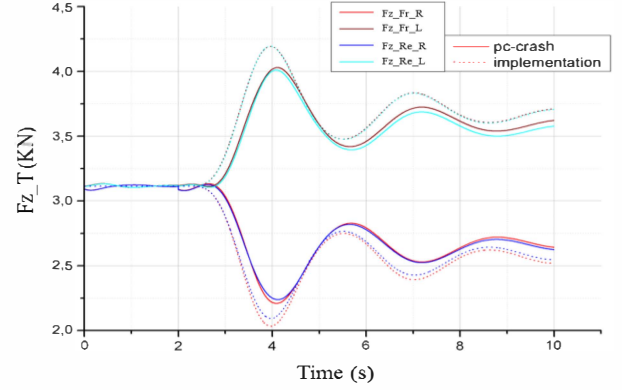


Fig. 7. Vertical Forces wheel / trailer

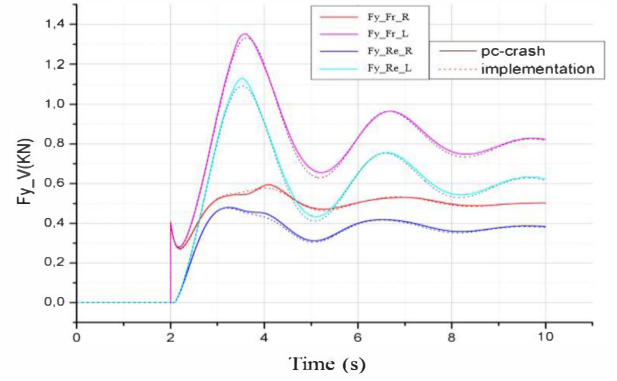


Fig. 8. Lateral Forces wheel / tractor

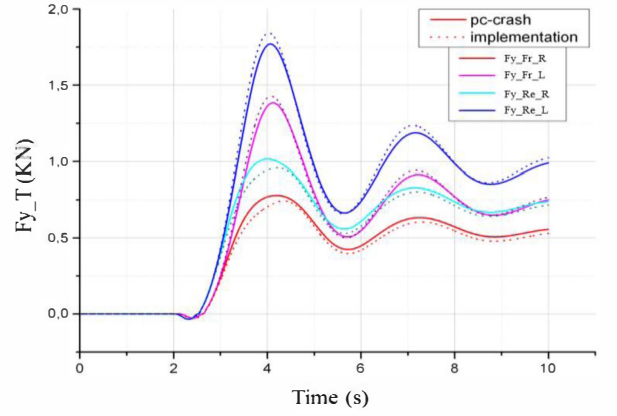


Fig. 9. Lateral Forces wheel / trailer

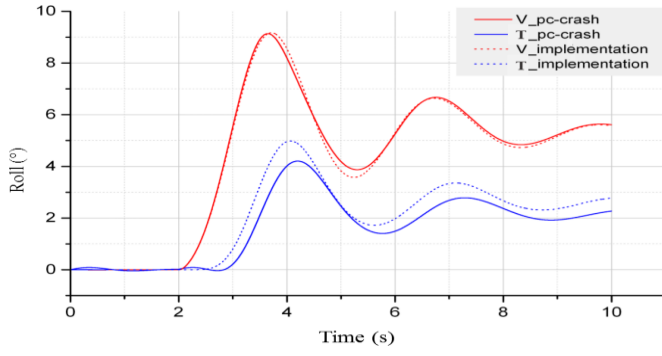


Fig. 10. Roll angle of the car-like robot

V. APPLICATION FOR UNBALANCED BRAKING IN A STRAIGHT LINE

For this scenario, our articulated car-like robot traveling at a speed of 90 km/h in a straight line, then applied a braking torque but assuming the braking distribution does not occur equally to the left and right of the same axle (this scenario can occur because of tire wear or difference in adherence to the roadway).

Figures show that case skid and deviation of up to 10 m on a 100 m race.

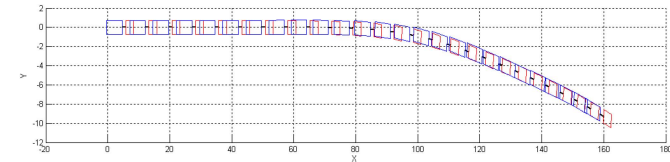


Fig. 11. Brake Difference 20% of the front right wheel relative to the left front wheel of the tractor and trailer braking

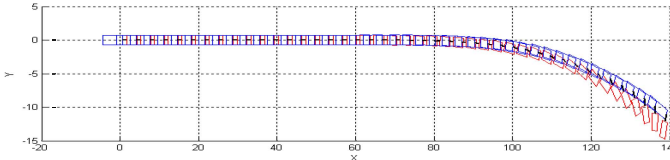


Fig. 12. Brake Difference 20% of the front right wheel relative to the left front wheel of the tractor without trailer braking.

The two cases shown in Figure 11 and 12 show the slippage due to the poor distribution of braking right and left, only for the front axle of the tractor. When the brake is applied the trailer was a drift but no rollover or jack-knife. By cons when you do not apply a brake on the trailer, it was the case of a jack-knife, and if we increase the braking torque distribution and the difference it will probably reversed.

VI. APPLICATION FOR JACK-KNIFE DUE TO BRAKING ON A BEND

In this case, we assume that our articulated vehicle takes the same turning (100 m) at a speed (15 m/s). Moreover, in the middle of bend is applied; in a first case a braking distributed on both vehicles and the second case is applied to the tractor braking only. In this, it is noted that the inertia of the trailer pushes the tractor until they form an angle of 90° there between; see figures 13 and 14.

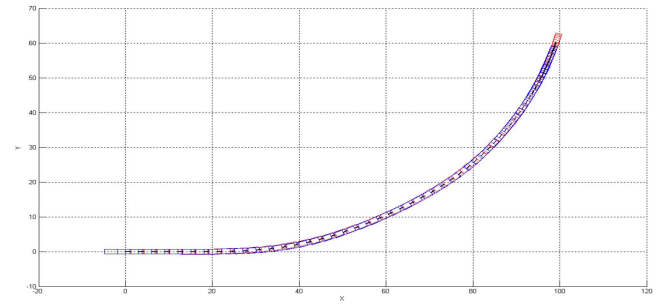


Fig. 13. Braking the tractor and trailer on a bend

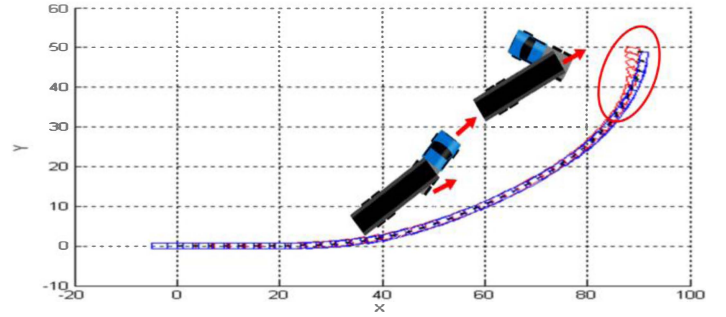


Fig. 14. Braking applied only to the tractor

VII. CONCLUSION

In this work, we detailed the dynamic model of a car-like robot with trailers. This model can be used to simulate various severe scenarios involving skidding overturning and jack-knifing. In the developed model, the wheel/ground interaction is modeled using a linear model and the link between the tractor and the trailer is handled as a spherical-joint. The simulation results obtained by the developed application are validated with those of the PC-Crash software. Comparison of results for accelerated and turned motion, revealed a very satisfactory correspondence, thus showing, consistency of the approach and the correctness of the implementation.

The simulations we have conducted have focused on the reproduction of normal operating conditions as well as loss of control situations such as lane departure, jack-knifing and overturning. As a perspective, control laws and mission planning will be studied in order to avoid such dangerous scenarios, involving skid and jack-knife phenomena.

REFERENCES

- [1] Steffan D. PC-Crash program for vehicle accidents. Technical manual of Pc-Crash, Linz, Austria, 2006.
- [2] Pacejka H. B., Bakker E. and Lidner L., A new tyer model with an application in vehicle dynamics studies. SAE International, paper 890087, 1989.
- [3] Reza N. Jazar, K. Vehicle dynamics. Theory and application. Ed. Springer, 2008.
- [4] Wolfgang H., User-appropriate tyer-modelling for vehicle dynamics in standard and limit situations, vehicle system dynamics. Swets and Zeitinger, vol. 38, pp. 103-125, 2002.
- [5] Datentechnik S., A simulation program for vehicle accidents. Technical manual of Pc-Crash, Linz, Austria, 2006.
- [6] B.HEIBING, M. ERSOY, Chassis Handbook. Fundamentals, Driving Dynamics, Components, Mechatronics, Perspectives, ATZ, Germany, 2011.